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Thin film capacitor, its manufacturing method, and electronic circuit substrate having the thin film capacitor

Abstract

A thin film capacitor includes: a metal foil having a roughened upper surface; a dielectric film covering the upper surface of the metal foil and having an opening through which the metal foil is partly exposed; a first electrode layer contacting the metal foil through the opening; a second electrode layer contacting the dielectric film without contacting the metal foil; and an insulating member separating the first and second electrode layers. The insulating member has a tapered shape in cross section. With the above configuration, both the first and second electrode layers can be disposed on the upper surface of the metal foil. In addition, since the insulating member has a tapered shape in cross section, adhesion performance of the insulating member can be enhanced, thus making it possible to prevent short-circuit between the first and second electrode layers.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is the U.S. National Phase under 35 U.S.C. § 371 of International Application No. PCT/JP2020/048393, filed on Dec. 24, 2020, which claims the benefit of U.S. Provisional Application No. 63/045,579, filed on Jun. 29, 2020, the entire contents of each are hereby incorporated by reference.

TECHNICAL FIELD
(1) The present invention relates to a thin film capacitor and its manufacturing method and, more particularly, to a thin film capacitor using a metal foil and its manufacturing method. The present invention also relates to an electronic circuit substrate having such a thin film capacitor.
BACKGROUND ART
(2) A decoupling capacitor is generally mounted on an IC-mounted circuit board so as to stabilize the potential of a power supply to be fed to the IC. A laminated ceramic chip capacitor is typically used as the decoupling capacitor, and a required decoupling capacitance is ensured by mounting many laminated ceramic chip capacitors on the surface of a circuit board.
(3) However, in recent years, a space for mounting many laminated ceramic chip capacitors is

sometimes insufficient due to miniaturization of circuit boards. Thus, a thin film capacitor capable of being embedded in a circuit board is sometimes used in place of the laminated ceramic chip capacitor (see Patent Documents 1 to 4).
(4) In the thin film capacitor described in Patent Document 1, a porous metal substrate is used, and an upper electrode is formed on the surface of the porous metal substrate through a dielectric film. In the thin film capacitor described in Patent Document 2, a metal substrate in which one main surface thereof is roughened is used, and an upper electrode is formed on the roughened surface of the metal substrate through a dielectric film. In the thin film capacitors described in Patent Documents 3 and 4, a conductive porous substrate is formed as a support part, and an upper electrode is formed on a roughened surface of the conductive porous substrate through a dielectric film.
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Patent Document
(5) [Patent Document 1] International Publication WO 2015/118901 [Patent Document 2] International Publication WO 2018/092722 [Patent Document 3] International Publication WO 2017/026247 [Patent Document 4] International Publication WO 2017/014020
SUMMARY OF THE INVENTION
Problem to be Solved by the Invention
(6) However, the thin film capacitor described in Patent Document 1 has a side surface electrode structure, so that the line length of the electrode is long, which causes a structural problem of increasing an ESR (Equivalent Series Resistance and an ESL (Equivalent Series Inductance). In addition, the thin film capacitor described in Patent Document 1 uses a metal substrate which is made entirely porous, so that it is not easy to separate the lower electrode constituted by the metal substrate and the upper electrode covering the metal substrate through a dielectric film, which disadvantageously makes it likely to cause a short circuit failure. In the thin film capacitor described in Patent Document 2, one main surface of the metal substrate functions as an upper electrode, and the other surface thereof functions as a lower electrode, so that it is necessary to route the electrode through the side surface of the element in order to dispose a pair of terminal electrodes on the same plane, complicating the structure. In the thin film capacitors described in Patent Documents 3 and 4, a pair of terminal electrodes are disposed on both surfaces of a metal substrate, respectively, preventing access to the terminal electrode pair from one side. In addition, the presence of the support increases the entire thickness.
(7) It is therefore an object of the present invention to provide an improved thin film capacitor and its manufacturing method. Another object thereof is to provide an electronic circuit substrate having such a thin film capacitor.
Means for Solving the Problem
(8) A thin film capacitor according to the present invention include: a metal foil having one roughened main surface, a dielectric film covering the one main surface of the metal foil and having an opening through which the metal foil is partly exposed, a first electrode layer contacting the metal foil through the opening, a second electrode layer contacting the dielectric film without contacting the metal foil, and a first insulating member separating the first and second electrode layers, wherein the first insulating member has a tapered shape in cross section such that the width thereof decreases as the distance from the one main surface of the metal foil increases.
(9) A manufacturing method for a thin film capacitor according to the present invention includes roughening one main surface of a metal foil, forming a dielectric film on the roughened one main surface of the metal foil, removing a part of the dielectric film to expose a part of the metal foil; forming, on the dielectric film, a first insulating member having a tapered shape such that the width thereof decreases as the distance from the one main surface of the metal foil increases; and forming a first electrode layer that contacts the part of the metal foil and a second electrode layer that contacts the dielectric film without contacting the part of the metal foil, the first and second

electrode layers being separated by the first insulating member.
Advantageous Effects of the Invention
(10) According to the present invention, an opening is formed in a part of the dielectric film, so that it is possible to dispose a pair of terminal electrodes on the same plane without using a side surface electrode or the like. In addition, since the first insulating member has a tapered shape in cross section, an adhesion performance of the first insulating member is enhanced, thereby a short-circuit between the first electrode layer and the second electrode layer can be prevented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a schematic cross-sectional view for explaining the structure of a thin film capacitor **1** according to an embodiment of the present invention.
(2) FIG. **1**B is a schematic plan view of the thin film capacitor **1**.
(3) FIG. **1**C is a schematic cross-sectional view illustrating an example in which a conductive member **32** is omitted from the thin film capacitor **1**.
(4) FIG. **1**D is a schematic plan view of the thin film capacitor **1** illustrated in FIG. **1**C.
(5) FIG. **2** is a schematic cross-sectional view for explaining the shape of a side surface **13** of the thin film capacitor **1**.
(6) FIG. **3** is a process view for explaining a manufacturing method for the thin film capacitor **1**.
(7) FIG. **4** is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(8) FIG. **5**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(9) FIG. **5**B is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(10) FIG. **6** is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(11) FIG. **7**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(12) FIG. **7**B is a schematic plan view of FIG. **7**A.
(13) FIG. **8**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(14) FIG. **8**B is a schematic plan view of FIG. **8**A.
(15) FIG. **9** is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(16) FIG. **10** is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(17) FIG. **11**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(18) FIG. **11**B is a schematic plan view of FIG. **11**A.
(19) FIG. **12**A is a schematic cross-sectional view illustrating an example of the formation position of an insulating member **21**.
(20) FIG. **12**B is a schematic cross-sectional view illustrating another example of the formation position of an insulating member **21**.
(21) FIG. **13**A is a schematic cross-sectional view for explaining the shape of the insulating member **21**.
(22) FIG. **13**B is a schematic cross-sectional view illustrating an example of the shape of the insulating member **21**.
(23) FIG. **13**C is a schematic cross-sectional view illustrating another example of the shape of the insulating member **21**.
(24) FIG. **14**A is a process view for explaining the manufacturing method for the thin film

capacitor **1**.
(25) FIG. **14**B is a schematic plan view of FIG. **14**A.
(26) FIG. **15**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(27) FIG. **15**B is a schematic plan view of FIG. **15**A.
(28) FIG. **16**A is a schematic cross-sectional view illustrating a case where the crystal particle diameter of a metal foil **10** is large.
(29) FIG. **16**B is a schematic plan view of FIG. **16**A.
(30) FIG. **17**A is a schematic cross-sectional view illustrating a case where the crystal particle diameter of a metal foil **10** is small.
(31) FIG. **17**B is a schematic plan view of FIG. **17**A.
(32) FIG. **18** is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(33) FIG. **19** is a schematic plan view of FIG. **18**.
(34) FIG. **20**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(35) FIG. **20**B is a schematic plan view of FIG. **20**A.
(36) FIG. **21** is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(37) FIG. **22**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(38) FIG. **22**B is a schematic plan view of FIG. **22**A.
(39) FIG. **23**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(40) FIG. **23**B is a schematic plan view of FIG. **23**A.
(41) FIG. **24**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(42) FIG. **24**B is a schematic plan view of FIG. **24**A.
(43) FIG. **25**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(44) FIG. **25**B is a schematic plan view of FIG. **25**A.
(45) FIG. **26**A is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(46) FIG. **26**B is a schematic plan view of FIG. **26**A.
(47) FIG. **27** is a process view for explaining the manufacturing method for the thin film capacitor **1**.
(48) FIG. **28** is a schematic cross-sectional view illustrating an electronic circuit substrate having a configuration in which the thin film capacitor **1** is embedded in a multilayer substrate **100**.
(49) FIG. **29** is a schematic cross-sectional view illustrating an electronic circuit substrate having a configuration in which the thin film capacitor **1** is mounted on the surface of a multilayer substrate **300**.
(50) FIG. **30** is a table indicating an evaluation results of samples.
MODE FOR CARRYING OUT THE INVENTION
(51) Hereinafter, a preferred embodiment of the present invention will be described in detail with reference to the accompanying drawings.
(52) FIG. **1**A is a schematic cross-sectional view for explaining the structure of a thin film capacitor **1** according to an embodiment of the present invention. FIG. **1**B is a schematic plan view of the thin film capacitor **1**.
(53) As illustrated in FIGS. **1**A and **1**B, the thin film capacitor **1** includes a metal foil **10**, ring-shaped or polygonal annular insulating members **21** and **22** formed on an upper surface **11** of the

metal foil **10**, conductive members **31** and **32** formed on the upper surface **11** of the metal foil **10** and partitioned by the insulating members **21** and **22**, a terminal electrode **51** connected to the conductive member **31** through a seed layer **40**, and a terminal electrode **52** connected to the conductive member **32** through the seed layer **40**. The metal foil **10** is made of a metal material such as aluminum, copper, chrome, nickel, or tantalum, and upper and lower surfaces **11** and **12** thereof positioned on the mutually opposite sides are each at least partly roughened. Aluminum is most preferable as the material of the metal foil **10**. A dielectric film D is formed on the upper and lower surfaces **11** and **12** of the metal foil **10**. The insulating members **21** and **22** are each made of, for example, resin. The conductive members **31** and **32** are each made of, for example, a conductive polymer material. The seed layer **40** and terminal electrodes **51** and **52** are each made of, for example, copper, nickel, or gold, an alloy thereof, or a layer structure thereof.

(54) The ring-shaped or polygonal annular insulating member **21** is provided in a slit that electrically isolating an electrode layer including the terminal electrode **51** and conductive member **31** from an electrode layer including the terminal electrode **52** and conductive member **32**. The terminal electrode **52** and conductive member **32** are positioned within an area surrounded by the insulating member **21**, and the terminal electrode **51** and conductive member **31** are positioned outside the area surrounded by the insulating member **21** and within an area surrounded by the insulating member **22**. In the area surrounded by the insulating member **21**, a part of or the entire dielectric film D formed on the upper surface **11** of the metal foil **10** is removed, and thus an opening is formed in the dielectric film D. As a result, the terminal electrode **52** is electrically connected to the metal foil **10** through the conductive member **32**. Alternatively, as illustrated in FIGS. **1**C and **1**D, the conductive member **32** may be omitted to connect the metal foil **10** and terminal electrode **52** directly or through the seed layer **40**. On the other hand, at the outside of the area surrounded by the insulating member **21**, the dielectric film D formed on the upper surface **11** of the metal foil **10** is not removed. That is, the conductive member **31** contacts the dielectric film D without contacting the metal foil **10**, so that the terminal electrode **51** and metal foil **10** are electrically isolated from each other. Thus, the terminal electrodes **51** and **52** function as a pair of capacitance electrodes opposite to each other through the dielectric film D. The dielectric film D is formed on the roughened upper surface **11** of the metal foil **10**, that is, the surface area of the upper surface **11** is increased, allowing a large capacitance to be obtained.

(55) As illustrated in FIG. **13**A, the insulating member **21** has a tapered shape in cross section such that a width W decreases as the distance from the upper surface **11** of the metal foil **10** increases. The cross section of the insulating member **21** is a cross section perpendicular to the extending direction of the insulating member **21**, which corresponds to the radial cross section of the ring-shaped insulating member **21**. Since the insulating member **21** has such a tapered shape, adhesion performance of the insulating member **21** with respect to the dielectric film D and metal foil **10** is enhanced to make short-circuit between the terminal electrodes **51** and **52** due to peeling of the insulating member **21** less likely to occur. In addition, the width of the insulating member **21** is small at its upper portion, so that a gap between the terminal electrodes **51** and **52** can be reduced. This allows the thin film capacitor **1** to be embedded in a fine-pitch circuit substrate. A width W**1** of the lower surface of the tapered insulating member **21** refers to the linear distance of a part of the insulating member **21** that contacts the dielectric film D and metal foil **10** in the width direction, and a width W**2** of the upper surface of the tapered insulating member **21** refers to the linear distance of the upper surface of the insulating member **21** in the width direction. When the upper surface of the insulating member **21** is not flat, the width at a 90% height position is defined as the width W**2**. That is, assuming that the height of the insulating member **21** is H, the width of the insulating member **21** at a 9H/10 position is defined as the width w**2**. In the present embodiment, W**1**>W**2** is satisfied. Like the insulating member **21**, the insulating member **22** has a tapered shape in cross section.

(56) At the outside of the area surrounded by the insulating member **22**, the dielectric film D

formed on the upper surface **11** of the metal foil **10** is exposed. The roughened surface is thus exposed at the outer peripheral portion of the thin film capacitor **1**, so that adhesion when the thin film capacitor **1** is embedded in a multilayer substrate can be enhanced. A side surface **13** of the metal foil **10** is not roughened and is covered with an insulating film **14**. The ring-shaped or polygonal annular insulating member **22** exists between the conductive member **31** and the side surface **13** of the metal foil **10**, and a clearance area where no conductive member exists is provided at the outside of the ring-shaped or polygonal annular insulating member **22**, so that even when the insulating film **14** is thin, a short circuit between the conductive member **31** and metal foil **10** is prevented.
(57) The crystal particle diameter of the center portion (non-roughened portion) of the metal foil **10** is preferably less than 15 μm in the planar direction (direction parallel to the upper and lower surfaces **11** and **12**) and less than 5 μm in the thickness direction (direction orthogonal to the upper and lower surfaces **11** and **12**), and crystal orientations are preferably aligned with the planar direction as much as possible. This can enhance the positional accuracy of the side surface **13** as will be described later.
(58) The thin film capacitor **1** can be used as a decoupling capacitor when being embedded in a multilayer substrate. The thickness of the thin film capacitor **1** is as very thin as, for example, 50 μm or less. Thus, when the terminal electrode **51** and conductive member **31** are formed on the upper surface **11** side, the thin film capacitor **1** is more likely to protrude toward the lower surface **12** side. Therefore, in order to suppress warpage of the element when it is embedded in the multilayer substrate, the following configuration is preferable. That is, when a straight line L**1** extending along the upper surface **11**, a straight line L**2** extending along the lower surface **12**, and a straight line L**3** extending along the side surface **13** are defined in the cross section illustrated in FIG. **2**, an angle θa formed by the straight lines L**2** and L**3** is preferably more than 20° and less than 80° ($20° < θa < 80°$). This means that the lower surface **12** preferably has a larger area than the upper surface **11**. This improves adhesion between the side surface **13** of the thin film capacitor **1** and the multilayer substrate, thus making it possible to increase strength and reliability of the thin film capacitor **1**. In this case, $30° \leq θa \leq 60°$ is more preferably satisfied. When the angle θa is designed to fall within the above range, warpage of the thin film capacitor **1** at the time of mounting is reduced, and a contact area between the side surface **13** and the insulating resin constituting the multilayer substrate is controlled optimally, so that strength and reliability of the thin film capacitor **1** can be further improved. The side surface **13** may be curved such that the angle θa becomes larger toward the upper surface **11** and becomes smaller toward the lower surface **12**. When the angle θa is not constant as just described, the value of the angle θa is defined by an average value.
(59) The following describes an example of a manufacturing method for the thin film capacitor **1**.
(60) First, the metal foil **10** made of aluminum with a thickness of about 50 μm is prepared (FIG. **3**), and the upper and lower surfaces **11** and **12** are each etched for roughening (FIG. **4**). In place of roughening the flat metal foil **10**, the metal foil **10** may be formed by sintering metal powder. As a result, there are formed, in the metal foil **10**, a porous layer **11***a* positioned on the upper surface **11** side and a porous layer **12***a* positioned on the lower surface **12** side. A non-porous layer **10***a*, which is not subjected to roughening, is positioned between the porous layers **11***a* and **12***a*. At this time, it is sufficient to apply roughening to at least the upper surface **11**, and the lower surface **12** may not necessarily be roughened; however, roughening both the upper and lower surfaces **11** and **12** can prevent warpage of the metal foil **10**. The upper surface **11** is preferably etched under a condition that the surface area thereof is increased as much as possible. When both the upper and lower surfaces **11** and **12** are roughened, they may be etched under different etching conditions. For example, the lower surface **12** may be roughened under a condition that adhesion with respect to the multilayer substrate is enhanced as much as possible.
(61) Then, the dielectric film D is formed on the surface of the metal foil **10** (FIG. **5**A). The dielectric film D may be formed through oxidation of the metal foil **10** or using a film forming

method excellent in coverage performance, such as an ALD method, a CVD method, or a mist CVD method. Examples of the material of the dielectric film D include Al.sub.2O.sub.3, TiO.sub.2, Ta.sub.2O.sub.5, and SiNx. Amorphous may be used as the material of the dielectric film D. In this case, the composition ratio of the dielectric film D is not always the same as the composition ratio described above. At this time, it is sufficient to form the dielectric film D on at least the upper surface **11**, and the dielectric film D may not necessarily be formed on the lower surface **12**; however, forming the dielectric film D also on the lower surface **12** can provide an insulating property to the lower surface **12**. As illustrated in FIG. **5**B, the dielectric film D formed on the lower surface **12** may have the same composition ratio as the dielectric film D formed on the upper surface **11**, may be a barrier film E having a different composition ratio, and may be a laminated structure of the dielectric film D and barrier film E. When the barrier film E exists on the lower surface **12** of the metal foil **10**, it is possible to suppress intrusion of reaction product gas generated from resin constituting the multilayer substrate at the time of curing the multilayer substrate. After formation of the dielectric film D or barrier film E, a base material **60** for conveyance is stuck to the lower surface **12** of the metal foil **10** (FIG. **6**).
(62) Then, a photosensitive resist is formed on the upper surface **11** of the metal foil **10**, followed by exposure and development, to form a patterned resist **61** (FIGS. **7**A and **7**B). The resist **61** has an opening **62** for exposing the dielectric film D therethrough. The resist may be a positive type or a negative type.
(63) Then, a part of or the entire dielectric film D is removed with the resist **61** used as a mask to expose the metal foil **10** through the opening **62** (FIGS. **8**A and **8**B). The dielectric film D can be removed using a reverse sputtering method, an ion milling method, a RIE method, a wet etching method, or the like. At this stage, the upper surface **11** of the metal foil **10** has already been roughened, so that by using the reverse sputtering method, ion milling method, RIE method or the like, it is possible to prevent spread of an etchant due to a capillary phenomenon. However, a liquid etchant may be used in this process. Although the surface of the exposed metal foil **10** and the dielectric film D constitute substantially the same plane in the example illustrated in FIG. **8**A, the roughened metal foil **10** can protrude from the dielectric film D in some etching condition as illustrated in FIG. **9**.
(64) Then, after removal of the resist **61** (FIG. **10**), the insulating members **21** and **22** are formed on the upper surface **11** of the metal foil **10** (FIGS. **11**A and **11**B). The insulating members **21** and **22** can be formed by a photolithography patterning method, a screen printing method, a gravure printing method, an inkjet method, or the like. As a result, the insulating members **21** and **22** each have a tapered cross section as illustrated in FIG. **11**A. The insulating member **21** may overlap (FIG. **12**A) or may not overlap (FIG. **12**B) a portion at which the metal foil **10** is exposed. Further, the insulating members **21** and **22** each may not necessarily have a symmetric cross section and, as illustrated in FIG. **13**B, an angle θc of the outer part of the ring with respect to a center line C in the thickness direction of the metal foil **10** may be made smaller than an angle θb of the inner part of the ring with respect to the center line C to make wider the taper surface at the ring outer part than the taper surface at the ring inner part. This increases the area surrounded by the insulating member **21**, whereby an ESR decreases due to a reduction in contact resistance. Alternatively, as illustrated in FIG. **13**C, the angle θc of the outer part of the ring with respect to a center line C in the thickness direction of the metal foil **10** may be made larger than the angle θb of the inner part of the ring with respect to the center line C to make narrower the taper surface at the ring outer part than the taper surface at the ring inner part. This increases an area outside the area surrounded by the insulating member **21**, whereby a capacitance increases. The insulating member **21** contacts the conductive member **32** or terminal electrode **52** at the inner side surface thereof and contacts the conductive member **31** or terminal electrode **51** at the outer side surface thereof. With the above structure, abnormal stress is not generated during a contraction process in the formation of the insulating members **21** and **22**, making it possible to reduce occurrence of cracks at the roughened portion to

thereby improve manufacturing yield.
(65) Then, a photosensitive resist is formed on the upper surface **11** of the metal foil **10**, followed by exposure and development, to form a patterned resist **64** (FIGS. **14**A and **14**B). The resist **64** has an opening **65** for exposing therethrough an area outside the insulating member **22**. The resist may be a positive type or a negative type.
(66) Then, the metal foil **10** is removed with the resist **64** used as a mask to individualize the metal foil **10** (FIGS. **15**A and **15**B). The metal foil **10** can be removed by a wet etching method using an etchant such as acid. In this case, even when a liquid etchant is used, it does not spread beyond the insulating member **22**.
(67) To individualize the metal foil **10** more accurately, the crystal particle diameter of the center portion (non-roughened portion) of the metal foil **10** is preferably less than 15 μm in the planar direction and less than 5 μm in the thickness direction, as described above. When the crystal particle diameter of the metal foil **10** is 15 μm or more in the planar direction and 5 μm or more in the thickness direction, crystal particles protrude from the inner wall of the side surface **13** as illustrated in FIGS. **16**A and **16**B to increase a variation in the size of the individualized metal foil **10**. On the other hand, when the crystal particle diameter of the metal foil **10** is less than 15 μm in the planar direction and less than 5 μm in the thickness direction, the crystal particles appearing at the side surface **13** are small in size as illustrated in FIGS. **17**A and **17**B, thereby reducing variation in the size of the individualized metal foil **10**.
(68) Then, after removal of the resist **64** (FIGS. **18** and **19**), the conductive members **31** and **32** in a paste or liquid form each made of a conductive polymer material are formed in an area surrounded by the insulating member **22** (FIGS. **20**A and **20**B). Specifically, the conductive member **32** is positioned within an area surrounded by the insulating member **21**, and the conductive member **31** is positioned outside the area surrounded by the insulating member **21** and within an area surrounded by the insulating member **22**. The conductive members **31** and **32** are in a paste or liquid form and are thus filled up to the bottom of the porous layer **11***a* due to capillary action. As a result, the conductive member **31** contacts the dielectric film D without contacting the metal foil **10**, while the conductive member **32** contacts the metal foil **10**. Note that the terminal electrode **52** may be directly formed without forming the conductive member **32**.
(69) Then, the seed layer **40** is formed on the entire surface (FIG. **21**). The seed layer **40** can be formed using a sputtering method or the like. Subsequently, a photosensitive resist is formed on the upper surface **11** of the metal foil **10**, followed by exposure and development, to form a patterned resist **67** (FIGS. **22**A and **22**B). The resist **67** has openings **68** and **69**. The opening **68** is positioned within the area surrounded by the insulating member **22** and outside the area surrounded by the insulating member **21**. The opening **69** is positioned within the area surrounded by the insulating member **21**. As a result, a part of the seed layer **40** that covers the conductive member **31** is exposed through the opening **68**, and a part thereof that covers the conductive member **32** is exposed through the opening **69**. The resist may be a positive type or a negative type.
(70) In this state, electrolytic plating is performed to form the terminal electrodes **51** and **52** (FIGS. **23**A and **23**B). Then, after removing the resist **67** by ashing or the like (FIGS. **24**A and **24**B), the seed layer **40** is removed (FIGS. **25**A and **25**B). After that, the insulating film **14** is formed on the side surface **13** of the metal foil **10** (FIG. **26**), and the base material **60** for conveyance is removed by peeling or etching (FIG. **27**), whereby the thin film capacitor **1** illustrated in FIGS. **1**A and **1**B is completed. The insulating film **14** can be formed by oxidizing the side surface **13** of the metal foil **10** by an ashing process for removing the resist **67** or a heat treatment process. The terminal electrodes **51** and **52** may each be formed in plural numbers, and at least one pair of the terminal electrodes **51** and **52** will suffice.
(71) The thin film capacitor **1** according to the present embodiment may be embedded in a multilayer substrate **100** as illustrated in FIG. **28** or may be mounted on the surface of a multilayer substrate **300** as illustrated in FIG. **29**.

(72) An electric circuit substrate illustrated in FIG. **28** has a configuration in which a semiconductor IC **200** is mounted on the multilayer substrate **100**. The multilayer substrate **100** includes a plurality of insulating layers including insulating layers **101** to **104** and a plurality of wiring patterns including wiring patterns **111** and **112**. The number of insulating layers is not particularly limited. In the example illustrated in FIG. **28**, the thin film capacitor **1** is embedded between the insulating layers **102** and **103**. There are provided on the surface of the multilayer substrate **100** a plurality of land patterns including land patterns **141** and **142**. The semiconductor IC **200** has a plurality of pad electrodes including pad electrodes **201** and **202**. For example, one of the pad electrodes **201** and **202** is a power supply terminal, and the other one thereof is a ground terminal. The pad electrode **201** and land pattern **141** are connected to each other through a solder **211**, and the pad electrode **202** and the land pattern **142** are connected to each other through a solder **212**. The land pattern **141** is connected to the terminal electrode **51** of the thin film capacitor **1** through a via conductor **121**, the wiring pattern **111**, and a via conductor **131**. The land pattern **142** is connected to the terminal electrode **52** of the thin film capacitor **1** through a via conductor **122**, the wiring pattern **112**, and a via conductor **132**. With this configuration, the thin film capacitor **1** functions as a decoupling capacitor for the semiconductor IC **200**.
(73) An electric circuit substrate illustrated in FIG. **29** has a configuration in which a semiconductor IC **400** is mounted on the multilayer substrate **300**. The multilayer substrate **300** includes a plurality of insulating layers including insulating layers **301** and **302** and a plurality of wiring patterns including wiring patterns **311** and **312**. The number of insulating layers is not particularly limited. In the example illustrated in FIG. **29**, the thin film capacitor **1** is surface-mounted on a surface **300***a* of the multilayer substrate **300**. There are provided on the surface **300***a* of the multilayer substrate **300** a plurality of land patterns including land patterns **341** to **344**. The semiconductor IC **400** has a plurality of pad electrodes including pad electrodes **401** and **402**. For example, one of the pad electrodes **401** and **402** is a power supply terminal, and the other one thereof is a ground terminal. The pad electrode **401** and the land pattern **341** are connected to each other through a solder **411**, and the pad electrode **402** and the land pattern **342** are connected to each other through a solder **412**. The land pattern **341** is connected to the terminal electrode **51** of the thin film capacitor **1** through a via conductor **321**, the wiring pattern **311**, a via conductor **331**, and a solder **413**. The land pattern **342** is connected to the terminal electrode **52** of the thin film capacitor **1** through a via conductor **322**, the wiring pattern **312**, the land pattern **344**, and a via conductor **332**, and a solder **414**. With this configuration, the thin film capacitor **1** functions as a decoupling capacitor for the semiconductor IC **400**.
(74) While the preferred embodiment of the present invention has been described, the present invention is not limited to the above embodiment, and various modifications may be made within the scope of the present invention, and all such modifications are included in the present invention.
EXAMPLES
(75) Samples of a plurality of thin film capacitors having the same configuration as that of the thin film capacitor **1** illustrated in FIG. **1** were prepared with the cross-sectional shape of the insulating member **21** varied from one sample to another. Each thin film capacitor was mounted in an evaluation multilayer substrate and left standing in a cooling atmosphere at −55° C. for 30 minutes and in a heating atmosphere at 150° C. for 30 minutes (cooling-heating cycle test), and a short-circuit failure rate was evaluated. The results are shown in FIG. **30**. In FIG. **30**, the “taper lower surface width” corresponds to the W**1** illustrated in FIG. **13**A, and the “taper upper surface width” corresponds to the W**2** illustrated in FIG. **13**A. Further, when the difference between θb and θc is within 2°, the insulating member **21** is regarded to have a symmetric cross section in which θb and θc are substantially the same.
(76) As shown in FIG. **30**, of samples A**1** to A**3** and B**1** in which θb and θc are substantially the same, samples A**1** to A**3** in which θb and θc are each less than 90°, that is, the insulating member **21** has a tapered shape, the occurrence probability of the short-circuit failure was 8% to 16%, while

in sample B**1** in which θb and θc are 90°, the occurrence probability of the short-circuit failure was 35%. In samples A**4** to A**9** in which θb≠θc, the occurrence probability of the short-circuit failure was 5% or less. In particular, in samples A**4** to A**6** in which θb>θc, the ESR was low.
REFERENCE SIGNS LIST
(77) **1**: Thin film capacitor **10**: Metal foil **10***a*: Non-porous layer **11**: Upper surface of metal foil **11***a*: Porous layer **12**: Lower surface of metal foil **12***a*: Porous layer **13**: Side surface of metal foil **14**: Insulating film **21**, **22**: Insulating member **31**, **32**: Conductive member **40**: Seed layer **51**, **52**: Terminal electrode **60**: Base material for conveyance **61**, **64**, **67**: Resist **62**, **65**, **68**, **69**: Opening **100**, **300**: Multilayer substrate **101** to **104**, **301**, **302**: Insulating layer **111**, **112**, **311**, **312**: Wiring pattern **121**, **122**, **131**, **132**, **321**, **322**, **331**, **332**: Via conductor **141**, **142**, **341** to **344**: Land pattern **200**, **400**: Semiconductor IC **201**, **202**, **401**, **402**: Pad electrode **211**, **212**, **411** to **414**: Solder **300***a*: Surface of multilayer substrate D: Dielectric film E: Barrier film

Claims

1. A thin film capacitor comprising: a metal foil having one roughened main surface; a dielectric film covering the one main surface of the metal foil and having an opening through which the metal foil is partly exposed; a first electrode layer contacting the metal foil through the opening; a second electrode layer contacting the dielectric film without contacting the metal foil; and a first insulating member separating the first and second electrode layers, wherein the first insulating member has a tapered shape in cross section such that a width thereof decreases as a distance from the one main surface of the metal foil increases.
2. The thin film capacitor as claimed in claim 1, wherein the first and second electrode layers are separated from each other by an annular slit, wherein the first electrode layer is provided in a first area surrounded by the slit, wherein the second electrode layer is provided in a second area positioned outside the slit, and the first insulating member is provided inside the slit.
3. The thin film capacitor as claimed in claim 2, wherein the first insulating member has a first side surface contacting the first electrode layer and a second side surface contacting the second electrode layer, and wherein, assuming that an angle formed by the one main surface of the electrode layer and the first side surface of the first insulating member is θb and that an angle formed by the one main surface of the electrode layer and the second side surface of the first insulating member is θc, both the θb and θc are less than 90°.
4. The thin film capacitor as claimed in claim 3, wherein the θb and θc are equal to each other.
5. The thin film capacitor as claimed in claim 3, wherein the θb and θc differ from each other.
6. The thin film capacitor as claimed in claim 5, wherein the θb is larger than the θc.
7. The thin film capacitor as claimed in claim 5, wherein the θb is smaller than the θc.
8. The thin film capacitor as claimed in claim 2, further comprising a second insulating member provided on the one main surface of the metal foil and surrounding the second electrode layer.
9. The thin film capacitor as claimed in claim 8, wherein the second insulating member has a tapered shape in cross section such that a width thereof decreases as a distance from the one main surface of the metal foil increases.
10. The thin film capacitor as claimed in claim 1, wherein other main surface of the metal foil is roughened.
11. The thin film capacitor as claimed in claim 1, wherein the second electrode layer includes a first conductive member contacting the dielectric film and made of a conductive polymer material and a second conductive member contacting the first conductive member and made of a metal material.
12. The thin film capacitor as claimed in claim 11, wherein the first electrode layer includes a third conductive member contacting the metal foil and made of a conductive polymer material and a fourth conductive member contacting the third conductive member and made of a metal material.
13. The thin film capacitor as claimed in claim 11, wherein the first electrode layer includes a

fourth conductive member contacting the metal foil and made of a metal material.

14. An electronic circuit substrate comprising: a substrate having a wiring pattern; a semiconductor IC provided in the substrate; and the thin film capacitor as claimed in claim 1 provided in the substrate, wherein the first and second electrode layers of the thin film capacitor are connected to the semiconductor IC through the wiring pattern.

15. A manufacturing method for a thin film capacitor, the method comprising: roughening one main surface of a metal foil; forming a dielectric film on the roughened one main surface of the metal foil; removing a part of the dielectric film to expose a part of the metal foil; forming, on the dielectric film, a first insulating member having a tapered shape such that a width thereof decreases as a distance from the one main surface of the metal foil increases; and forming a first electrode layer that contacts the part of the metal foil and a second electrode layer that contacts the dielectric film without contacting the part of the metal foil, the first and second electrode layers being separated by the first insulating member.

16. The manufacturing method as claimed in claim 15, wherein the first electrode layer is formed in an area surrounded by the first insulating member, and wherein the second electrode layer is formed in an area outside the first insulating member.

17. The manufacturing method as claimed in claim 16, further comprising forming a second insulating member that surrounds the first insulating member, wherein, after the forming the first and second insulating members, the second electrode layer is formed in an area surrounded by the second insulating member.

18. The manufacturing method as claimed in claim 15, wherein the second electrode layer is formed by forming a first conductive member in a paste or liquid form on the dielectric film and forming a second conductive member made of a metal material on a surface of the first conductive member.

19. The manufacturing method as claimed in claim 15, further comprising roughening other main surface of the metal foil.